

**PROFESSIONAL SUMMARY:**

- 10+ years of experience in AI/ML and Data Science, architecting and deploying production-grade machine learning systems that drive data-driven products and strategic business decisions at scale across enterprise environments.
  - Proven expertise in delivering end-to-end ML solutions spanning data engineering, feature development, model training, deployment, and lifecycle management in complex, real-world settings.
  - Strong foundation in classical machine learning and statistical modeling across text, tabular, and image data, including classification, regression, clustering, and time-series forecasting for business applications.
  - Skilled in feature engineering, model selection, and hyperparameter tuning with rigorous evaluation using cross-validation, F1-score, AUC-ROC, RMSE, and statistical testing to ensure robust and generalizable models.
  - Deep understanding of overfitting prevention, data leakage control, and reproducible experimentation practices to maintain model integrity and trust.
  - Hands-on experience in NLP and LLM workflows, including text preprocessing, TF-IDF vectorization, embeddings, dimensionality reduction (PCA), and structured evaluation pipelines.
  - Designed adversarial and edge-case testing strategies to analyze model robustness and failure modes, implementing rule-based fallback systems to improve production reliability.
  - Specialized in Large Language Models and Generative AI, building and deploying RAG pipelines, vector search systems, and prompt-engineered solutions for knowledge assistants, document Q&A, and intelligent automation.
  - Experienced in reducing hallucinations and improving grounding in LLM applications through evaluation pipelines, retrieval strategies, and practical guardrails.
  - Familiar with multi-agent and tool-augmented LLM systems, safe tool usage, and human-in-the-loop feedback loops for controlled and reliable AI behavior.
  - Experienced across the full ML lifecycle from data ingestion and EDA to deployment, monitoring, and iterative improvement in production.
  - Extensive experience building scalable data engineering pipelines on Azure Databricks and PySpark, processing multi-terabyte datasets for downstream ML training and real-time inference workflows.
  - Skilled in designing distributed ETL/ELT pipelines that ensure feature consistency between training and serving environments, reducing data quality issues and eliminating training-serving skew in production systems.
  - Proven ability to architect end-to-end data workflows—from ingestion and large-scale transformation in Spark to validated, analytics-ready datasets—supporting both machine learning models and BI reporting needs across regulated, high-volume environments.
  - Strong cloud experience with Azure, developing scalable ML systems, low-latency inference APIs, and both real-time and batch scoring pipelines.
  - Applied MLOps practices, including experiment tracking, CI/CD integration, model versioning, and monitoring to support stable production deployments.
  - Designed and implemented LLM evaluation and governance workflows, including hallucination tracking, retrieval quality validation, and structured output monitoring for production GenAI systems.
  - Built and deployed secure GenAI pipelines in regulated environments, ensuring controlled data access, safe output generation, and compliance with enterprise governance standards.
  - Optimized training and inference workflows for production AI systems by balancing latency, throughput, and cost across real-time APIs and scalable cloud infrastructure.
  - Delivered measurable business value through AI initiatives, including automation, cost reduction, risk mitigation, and operational efficiency improvements.
  - Designed responsible AI and model governance frameworks, incorporating fairness checks, explainability, human-in-the-loop validation, and audit-ready documentation aligned with enterprise risk standards.
  - Built LLM evaluation pipelines with structured benchmarking, including retrieval quality metrics, prompt evaluation, and controlled output validation to ensure reliability in production GenAI systems.
  - Effective collaborator with product managers, engineers, and business teams to define problems, align on metrics, and ship impactful solutions.
  - Strong problem-solving mindset, ownership attitude, and commitment to continuous learning in a rapidly evolving AI landscape.
  - Passionate about combining strong ML fundamentals with modern Generative AI to build practical, responsible, and scalable AI systems aligned with real business needs.
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TECHNICAL SKILLS:

| Category                          | Key Tools & Concepts                                                                                                                                                                                      |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Programming Languages             | Python, SQL                                                                                                                                                                                               |
| Machine Learning                  | Supervised & Unsupervised Learning, XGBoost, LightGBM, Random Forest, SVM, K-Means, DBSCAN, PCA, Ensemble Methods, Anomaly Detection, classification, regression, clustering                              |
| Deep Learning                     | TensorFlow, PyTorch, Keras, CNN, RNN, LSTM, Attention Mechanism, BERT, GPT                                                                                                                                |
| Natural Language Processing (NLP) | Hugging Face Transformers, spaCy, NLTK, LangChain, LlamaIndex, OpenAI GPT, Text Classification, NER, Summarization, Q&A Systems, Retrieval-Augmented Generation (RAG)                                     |
| Generative AI & LLMs              | Prompt Engineering, Fine-tuning LLMs, Synthetic Data Generation, LangChain, LangGraph, Crew AI, OpenAI APIs (ChatGPT, DALL-E), LLaMA, Claude (Anthropic)                                                  |
| Vector Databases                  | FAISS, Pinecone, Weaviate, ChromaDB, Elasticsearch (for hybrid search), Azure Cognitive Search                                                                                                            |
| Agents & Orchestration            | LangChain Agents, Crew AI, OpenAI Function Calling (FTK), Agentic Frameworks (LangGraph, A2A Google), Task Execution Trees (TETs)                                                                         |
| MLOps & Deployment                | MLflow, DVC, Docker, FastAPI, Flask, Streamlit, Azure ML, Jenkins, CI/CD Pipelines, Model Serving, API Performance Tuning, Blue-Green Deployments, A/B Testing, Inference Optimization, Observability     |
| Data Engineering                  | Azure Databricks, Apache Airflow, Pandas, NumPy, Spark, ETL/ELT Pipelines, PySpark, Distributed Data Processing, Streaming Pipelines, Data Cleaning, Feature Engineering, Feast (Feature Store)           |
| Cloud Platforms                   | Microsoft Azure (incl. Azure OpenAI), AWS (S3, EC2, Bedrock with LLaMA/Claude Models, SageMaker), GCP (BigQuery, Vertex AI)                                                                               |
| Databases                         | PostgreSQL, MySQL, MongoDB                                                                                                                                                                                |
| Monitoring & Security             | Prometheus, Grafana, Model Drift Detection, Data Encryption, OAuth2.0, GDPR/HIPAA Compliance, IAM, Responsible AI Governance, OpenTelemetry, Hallucination Monitoring, Retrieval Quality Monitoring, RBAC |
| Visualization                     | Matplotlib, Seaborn, Plotly, Power BI, Tableau                                                                                                                                                            |
| Version Control                   | Git, GitHub, GitLab                                                                                                                                                                                       |

EDUCATION:

- Master of Science in Computing and Data Science
- Master of Business Administration

PROFESSIONAL EXPERIENCE:

TD Bank, Ewing Township, NJ  
Senior Machine Learning / AI Engineer

Oct 2023 - PRESENT

Project: AI-Powered KYC Document Intelligence & Verification System

**Tech Stack:** Azure Databricks, PySpark, Python, TensorFlow, FastAPI, Docker, MLflow, Azure, OpenAI API (LangChain), PostgreSQL

**Project Overview:** Developed an end-to-end production-grade system for automated KYC document processing, leveraging computer vision and Generative AI to classify, validate, and extract insights from identity documents such as passports, driver’s licenses, and financial records.

Roles and Responsibilities:

- Architected and deployed a high-performance REST API using FastAPI and Docker, wrapping a fine-tuned MobileNetV2 backbone for document classification (IDs, checks, invoices), enabling real-time inference with sub-200ms latency for customer-facing banking applications.

- Integrated MLflow for experiment tracking and model versioning, establishing a centralized Model Registry to manage lifecycle transitions between Staging and Production, ensuring auditability and regulatory compliance in model updates.
- Designed a Retrieval-Augmented Generation (RAG) pipeline where classified document outputs and extracted entities were fed into an LLM via LangChain to generate contextual insights such as fraud risk scores, transaction explanations, and compliance summaries.
- Developed a robust automated data validation framework to handle noisy document inputs (blurred scans, skewed images, OCR inconsistencies, and out-of-distribution formats), significantly reducing production inference errors and manual intervention.
- Conducted staged fine-tuning and hyperparameter optimization (GridSearchCV) on document classification models, improving baseline accuracy while maintaining a lightweight architecture suitable for scalable cloud deployment.
- Implemented stratified evaluation frameworks using Macro F1-score and per-class precision/recall analysis to address class imbalance across diverse document types and ensure fair and consistent model performance.
- Built scalable data engineering pipelines on Azure Databricks using PySpark to ingest, clean, and transform high-volume KYC document metadata and extracted entities for downstream model training and RAG retrieval.
- Designed distributed PySpark workflows to process and validate large batches of identity documents in parallel, improving throughput and reducing processing latency during peak audit and transaction cycles.
- Leveraged Azure Databricks for scalable feature engineering and data quality validation, ensuring consistent, analytics-ready datasets that supported both fraud-detection model training and compliance reporting.
- Engineered a CI/CD pipeline automating unit testing for ML components and integration testing for APIs, enabling zero-downtime deployments through Blue-Green strategies in a highly regulated banking environment.
- Implemented Prometheus and Grafana dashboards to monitor real-time model performance, including prediction drift, latency, and error rates, enabling proactive retraining and incident mitigation.
- Optimized inference costs through model quantization (TFLite/OpenVINO), reducing model size and compute requirements while preserving accuracy for deployment in cost-sensitive production environments.
- Designed an asynchronous processing pipeline using Celery and Redis to handle high-resolution document uploads and OCR workloads, improving system throughput and preventing API timeouts under heavy load.
- Leveraged ChromaDB/Pinecone to store document embeddings, enabling semantic and visual similarity search for historical transaction comparison, fraud detection, and enhanced RAG-based reasoning.
- Implemented secure access management using IAM policies, RBAC, OAuth2, and secrets management, ensuring strict governance and controlled access to sensitive banking systems and data.
- Built observability frameworks using OpenTelemetry to track retrieval accuracy, token usage, system latency, and model outputs, supporting proactive monitoring and rapid incident response.
- Established artifact traceability through CI/CD pipelines using GitHub Actions and Azure DevOps, ensuring reproducibility, version control, and compliance with enterprise audit requirements.
- Fine-tuned small language models (SLMs) using LoRA and adapter techniques to generate low-latency, cost-efficient financial insights while maintaining high data quality and reliability.
- Implemented automated retraining workflows triggered by model drift detection and performance degradation, ensuring continuous model improvement and operational trust in production systems.
- Applied hybrid cloud strategies leveraging Azure OpenAI/Azure ML services to distribute workloads across regions, ensuring high availability, scalability, and cost optimization.
- Optimized production inference systems through API performance tuning, request batching, caching strategies, and latency-throughput trade-off optimization, ensuring scalable and cost-efficient real-time document processing.
- Developed and operationalized LLM evaluation and governance pipelines, including structured benchmarking, retrieval quality metrics, hallucination detection, bias analysis, and human-in-the-loop validation, ensuring audit-ready outputs and compliance with banking regulations.

**Evolent, Arlington, VA**  
**Senior Data Scientist / ML Engineer**

**Nov 2020 – Jun 2023**

#### **Project: Enterprise Phishing Intelligence & Automated Risk Mitigation**

**Tech Stack:** Python (Scikit-learn, XGBoost), Spark, SQL Server, MLflow, Airflow, SHAP, Docker, Tableau

**Project Overview:** Engineered a comprehensive predictive threat-detection ecosystem to safeguard organizational assets. Successfully shifted security operations from reactive pattern-matching to a proactive, ML-driven defense posture, achieving a reduction in breach incidents while maintaining a strictly controlled False Positive Rate (FPR) to protect employee productivity.

#### **Roles and Responsibilities:**

- Translated technical metrics into business outcomes by developing a Financial Loss Avoidance model; calibrated classification thresholds to balance the high cost of data breaches against the operational friction of false alarms.
- Optimized system performance for high-volume traffic by implementing a Cascading Inference Architecture, utilizing a lightweight heuristic filter for traffic and routing complex edge cases to a high-fidelity XGBoost ensemble.
- Solved for Concept Drift by engineering an automated feedback loop; integrated human-in-the-loop labels from security analysts into an Airflow-orchestrated retraining pipeline to counter evolving zero-day tactics.
- Developed a robust library of lexical and behavioral features, including URL Shannon Entropy, domain-age cross-referencing, and n-gram TF-IDF analysis to detect sophisticated homograph and obfuscation attacks.
- Integrated SHAP (SHapley Additive exPlanations) to provide transparent, feature-level justifications for model decisions, reducing the Mean Time to Remediation (MTTR) for security analysts by providing interpretable "Risk Indicators."
- Leveraged PySpark for distributed feature engineering and large-scale data processing, enabling scalable model training workflows on multi-terabyte datasets.
- Architected Apache Spark ETL pipelines to process multi-terabyte communication logs, ensuring real-time feature consistency through a centralized feature-store approach using SQL Server.
- Established a model monitoring framework using MLflow and custom health dashboards; tracked Prediction Probability Calibration and feature-drift metrics to ensure long-term model reliability and regulatory auditability.
- Implemented Platt Scaling and Isotonic Regression to calibrate model output probabilities, which was critical for automated blocking policies.
- Conducted Adversarial Attacks (e.g., character-level perturbations) during the QA phase to identify model vulnerabilities, subsequently hardening the pipeline by augmenting the training set with synthetic adversarial samples.
- Reduced cloud infrastructure overhead by implementing Horizontal Pod Autoscaling (HPA) within a containerized environment and utilizing spot instances for non-latency-sensitive batch retraining jobs.
- Engineered an upstream data-quality layer using Great Expectations to validate schema consistency and detect "null-rate" spikes in incoming logs, preventing "silent failures" in model predictions.
- Eliminated Training-Serving Skew by implementing a versioned feature store logic, ensuring that the exact feature definitions used during Spark-based training were mirrored in the real-time inference environment.
- Hardened the ML pipeline for SOC2/GDPR compliance by enforcing TLS 1.3 for data in transit and implementing fine-grained IAM roles for access to sensitive communication metadata in SQL Server.
- Authored standardized MLOps playbooks for the broader data team, defining protocols for model rollbacks, A/B testing canary deployments, and incident response for model performance degradation.
- Developed real-time data pipelines using Azure Event Hubs and Stream Analytics to support immediate anomaly alerts on incoming transactional and communication data streams.
- Processed and analyzed unstructured communication logs using Azure Databricks (Spark), enabling scalable batch processing and distributed computing for downstream threat detection.
- Created NLP pipelines using SpaCy, NLTK, and custom embeddings to support keyword extraction and document clustering across financial communication logs to identify obfuscation patterns.
- Fine-tuned BERT and DistilBERT models using HuggingFace Transformers for sentiment analysis and escalation detection within suspicious client interactions.
- Applied PCA and SHAP for feature selection and interpretability, aligning with rigorous model risk management (MRM) and internal audit compliance requirements.
- Maintained full lifecycle governance by generating audit-ready metadata within Azure Purview aligned with GDPR and internal financial security policies.
- Evaluated and translated advanced modeling techniques, calibration methods, and adversarial testing approaches into production-ready workflows, balancing model robustness, explainability, and operational constraints.
- Communicated model behavior, risk trade-offs, and performance findings to technical and non-technical stakeholders, helping align deployment decisions with business and operational objectives.
- Orchestrated model training and batch inference workflows using Azure Machine Learning (Azure ML) Pipelines, automating data preprocessing and evaluation steps.

**Sakav Technology, Pune, MH**  
**Associate Data Scientist**

**Jun 2014 - Feb 2016**

**Project: Geospatial Valuation Engine & Real Estate Risk Analytics**

**Tech Stack:** Python, Scikit-learn, XGBoost, CatBoost, SQL, Tableau, SHAP, Statsmodels

**Project Overview**

Developed a high-fidelity automated valuation model (AVM) to predict residential property values across diverse regional markets. Focused on engineering spatially aware features and ensuring model transparency for stakeholder decision-making, resulting in a reduction in mean absolute percentage error (MAPE) compared to baseline institutional models.

#### **Roles and Responsibilities:**

- Engineered interaction terms and geospatial aggregates (e.g., price-per-square-foot trends at the neighborhood level); utilized Recursive Feature Elimination (RFE) and Lasso (L1) regularization to identify the most predictive variables while reducing model complexity.
- Deployed a stacked ensemble of Gradient Boosted Trees (XGBoost/CatBoost) to capture complex non-linear relationships in housing data, significantly outperforming traditional OLS regression in high-volatility markets.
- Implemented a Time-Series Split validation strategy to prevent data leakage, ensuring the model could generalize to future market conditions rather than simply memorizing historical price points.
- Conducted rigorous Multicollinearity diagnostics (VIF) and Hypothesis Testing (p-values) on regression coefficients to ensure the structural integrity of the model and provide reliable feature-importance rankings.
- Integrated SHAP (SHapley Additive exPlanations) to decompose individual house price predictions into contributing factors (e.g., +\$10k for school district, -\$5k for age), providing "Reason Codes" essential for regulatory compliance.
- Designed an automated Anomaly Detection pipeline using Isolation Forests to flag unrealistic property listings or "fat-finger" data entry errors, ensuring data integrity before model training.
- Developed interactive Tableau Geospatial Dashboards that allowed stakeholders to drill down into localized market trends, residuals analysis, and "Predicted vs. Actual" performance by pin code.
- Performed residual analysis across different demographic and geographic slices to detect and mitigate valuation bias, ensuring the model performed equitably across underserved neighborhoods.
- Implemented Quantile Regression to quantify downside investment risk and provide localized "valuation bands" rather than static mean estimates.
- Leveraged Target Encoding with Prior Smoothing for pin codes and neighborhoods, capturing hyper-local market nuances while mitigating overfitting.
- Built time-series forecasting models using ARIMA/Prophet for trend analysis and future demand prediction
- Performed extensive EDA using Pandas, Matplotlib, and Seaborn to identify seasonal patterns and anomalies
- Developed dashboards in Power BI to visualize trends and support decision-making
- Engineered a Champion-Challenger (Shadow Mode) framework to validate model performance on live market data before full production promotion.
- Remediated Heteroscedasticity via log-transforms and Weighted Least Squares (WLS), ensuring reliable prediction intervals across luxury and entry-level segments.
- Built a Boruta-based selection pipeline to isolate signal from noise, socioeconomic indicators, and alternative datasets.
- Orchestrated SQL-based ingestion of alternative geospatial data (crime, transit, schools), driving significant model lift in urban sub-markets.
- Applied Post-Hoc Probability Calibration to align confidence scores with historical market outcomes, ensuring actionable output for stakeholders.
- Enterprise ETL Ingestion: Developed end-to-end ETL pipelines using Azure Data Factory (ADF) and Azure Data Lake Storage (ADLS Gen2) to process large-scale geospatial, CRM, and neighborhood engagement data.
- High-Performance Modeling: Built predictive models using XGBoost and LightGBM for valuation and customer lifetime value, utilizing Optuna and Bayesian optimization for hyperparameter tuning.
- Complex Feature Engineering: Engineered features including temporal lags and rolling aggregates using Featuretools and Pandas for time-series behavior modeling of localized market trends.
- Rare-Event Handling: Addressed class imbalance in rare-event prediction models using SMOTE and cost-sensitive learning, ensuring model robustness across diverse regional segments.
- Advanced Visualization: Designed integrated dashboards in Power BI, connecting to Azure Synapse Analytics to visualize KPIs, model outputs, and localized risk metrics.
- Worked with business stakeholders to translate valuation requirements into explainable modeling approaches and decision-support outputs aligned with operational goals.
- Conducted structured experimentation across feature sets, validation strategies, and model families, translating analytical research into deployable predictive solutions.
- Documented model assumptions, validation strategies, and explainability outputs as audit-ready artifacts, supporting transparency, reproducibility, and stakeholder trust in valuation decisions.
- Model Management: Leveraged Azure ML Model Registry to version models and track hyperparameter experiments, ensuring reproducibility across regional market deployments.

**Project: Enterprise Reporting & Database Support**

**Tech Stack:** SQL Server (T-SQL), SSIS (Integration Services), SSRS (Reporting Services), MS Excel, Python, Jira

**Project Overview**

Supported the development of a centralized reporting system for vehicle safety data. Focused on automating data entry processes and creating standardized reports to ensure high data quality and accuracy for executive decision-making.

**Roles and Responsibilities:**

- **SSIS Package Development:** Created and scheduled automated SSIS packages to move data from text files into SQL Server; moved the team from manual data loading to an automated daily refresh schedule.
  - **Data Reconciliation & Audit:** Conducted thorough Source-to-Target (S2T) audits using SQL comparison scripts; identified and corrected calculation errors that resulted in the recovery of misallocated costs.
  - **Standardized Reporting (SSRS):** Developed a library of automated reports in SSRS, allowing users to filter by date and region, which eliminated the need for manual one-off data requests.
  - **Excel Automation:** Used Python (Pandas) to clean and format raw data files before they were loaded into the database, reducing manual preparation hours per week.
  - **Data Quality Assurance:** Performed daily data profiling to check for missing values or duplicates, ensuring that all published figures were accurate.
  - **Requirement Documentation:** Met with business users to document what data they needed and translated those needs into technical "Data Dictionaries."
  - **Anomalies & Bug Tracking:** Investigated why numbers didn't match between systems, found the "root cause" in the source data, and logged the issues in Jira.
  - **User Testing (UAT):** Guided business users through "Acceptance Testing" to make sure the reports were easy to use and showed the correct numbers before final release.
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**Authorized to work in the USA (EAD holder). No visa sponsorship is required. Open to relocation.**

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**CERTIFICATIONS:**

- CompTIA IT Fundamental
- Scrum Master Certified